PROJECT REPORT: PHISHING URL DETECTION USING MACHINE LEARNING

1. INTRODUCTION

- As the number of internet users increases exponentially, the threats associated with internet usage are also rising. Phishing is one of the most common internet crimes where internet users are tricked into visiting fake websites through false links that resemble the original ones.
- **Problem Statement:** Phishing is a type of internet crime where internet users are tricked into visiting fake websites through false links that resemble the original ones. This is done in such a way that the internet user is unable to distinguish between the original and fake ones.

This project aims at developing a machine learning-based system that can be used to determine whether the given URL is legitimate or not.

2. OBJECTIVES

The objectives of the project are:-

- Identify common patterns in the URLs of the phishing sites,
- Convert the text data into a suitable format,
- Develop the system,
- Evaluate the system,
- Create a user-friendly interface.

3. METHOD

- Data Collection
- Feature Extraction
- Model Selection
- Model Training
- Evaluation

4. IMPLEMENTATION

- Implementation of the project is done in Python using the following libraries:

1. Pandas
2. scikit-learn

- Sample Dataset
- Data Frame
- Convert Text Into Features
- Training The Model
- Function To Predict If A URL Is Fake Or Not
- User Input

5. RESULTS

- - Recognizes common phishing signs appropriately,
- Works well if detection is dependent on key words,
- Demonstrates a good example of machine learning.

6. ADVANTAGES

- Simple and easy to understand
- Fast predictions
- Simple design
- Simple extension.

7. APPLICATIONS FOR THIS PROJECT

- Email spam detection systems
- Security tools built into browsers
- Cybersecurity systems
- User education systems

8. SUMMARY

- The goal of this project was to show how machine learning can be applied to the identification of phishing URLs based on their text-based attributes. Even with just a simple dataset and machine learning model, the detection of phishing URLs from either a machine learning perspective or an analyst's view demonstrated that there are clear patterns of suspicious activity that can be detected. As the tools and methods become more advanced with additional data sources, this tool can become a real-world cyber security solution.

MY CODE:

File Edit Selection View Go Run Terminal Help AIML PROJECT

EXPLORER

AIML ...

Phishing _URL.py

Phishing _URL.py > ...

```python
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB

# SAMPLE DATASET
data = {
    'text': [
        "http://free-money-now.com",
        "http://secure-bank-login.com",
        "Win cash prizes by clicking here!",
        "https://google.com",
        "claim your reward now",
        "https://github.com",
    ],
    "label": [1, 1, 1, 0, 1, 0]  # 1 = spam, 0 = safe
}

# CREATE DATAFRAME
df= pd.DataFrame(data)

# CONVERT TEXT INTO FEATURES
vectorizer = TfidfVectorizer()
x= vectorizer.fit_transform(df['text'])

# TRAINING THE MODEL
model = MultinomialNB()
model.fit(x, df['label'])

# FUNCTION TO PREDICT IF A URL IS FAKE OR NOT
def predict_link(link):
    url_vector = vectorizer.transform([link])
    prediction = model.predict(url_vector)[0]
    return "SPAM " if prediction == 1 else "Probably Safe"

# USER INPUT
print(" 🔍 Link Spam Detector (type 'exit' to quit)\n")

while True:
    user_input = input("Enter a URL to check (like: http://free-money-now.com):  ")

    if user_input.lower() == 'exit':
        print("Exiting the Link Spam Detector. Stay safe online! 👋")
        break

    result = predict_link(user_input)
    print(f"Result: {result}\n")
```

OUTLINE

TIMELINE

0 0 Ln 47, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.14 (64-bit) Go Live

OUTPUT:

File Edit Selection View Go Run Terminal Help AIML PROJECT

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```

PROBLEMS OUTPUT TERMINAL PORTS

```
Enter a URL to check (like: http://free-money-now.com):  https://www.google.com/
Result: Probably Safe

Enter a URL to check (like: http://free-money-now.com):  http://paypal-support.ga
Result: SPAM

Enter a URL to check (like: http://free-money-now.com):  exit
Exiting the Link Spam Detector. Stay safe online! 👋
PS C:\Users\Avni\OneDrive\Documents\AIML PROJECT>
```

PREPARED BY:
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Phishing URL Detection using Machine Learning.